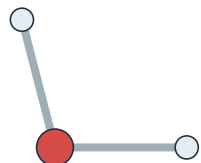


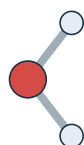
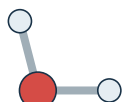
Water clusters | Python vs LAMMPS

Five isolated, neutral, unoptimized geometries · single-point energies and forces

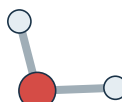
Monomer



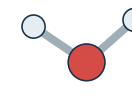
Dimer



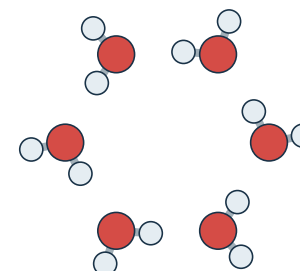
Distorted Dimer



Trimer



Hexamer



TOTAL ENERGIES / kcal/mol; $\Delta E = \text{Python} - \text{LAMMPS}$

Structure	Atoms	Python total	LAMMPS total	ΔE
monomer	3	-248.344209206	-248.344209206	+1.421e-13
dimer	6	-502.630151521	-502.630151521	+4.547e-13
distorted dimer	6	-498.807203052	-498.807203052	+2.842e-13
trimer	9	-756.928700873	-756.928700873	+4.547e-13
hexamer	18	-1524.961058062	-1524.961058062	-6.821e-13

FORCE DIFFERENCES / kcal/mol/Å; Cartesian components

Structure	Fixed-charge max $ \Delta F $	Fixed-charge RMS ΔF	Charge-response max $ \Delta F $
monomer	4.441e-14	2.208e-14	1.739e-01
dimer	7.603e-13	2.832e-13	1.841e-01
distorted dimer	4.294e-13	1.615e-13	1.747e-01
trimer	1.403e-13	4.428e-14	2.164e-01
hexamer	8.151e-12	2.566e-12	1.895e-01

Fixed-charge forces: QEq at each structure; charges held constant only during differentiation (default).

Charge-response forces: full derivative through QEq (optional). All differences subtract LAMMPS forces.

RMS averages all 3N components. Force conventions differ because $14.4 \times 23.02 \neq 332.06371$ in the reference.

Reference: LAMMPS (22 Jul 2025 - Update 4) via Imp_mpi. Parameters: qeq_ff.water; Achtyl et al. (2015).

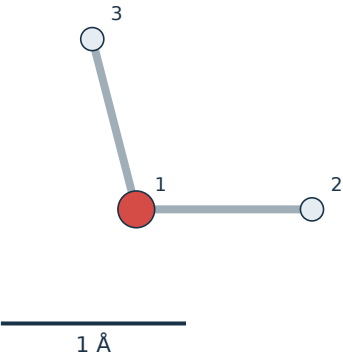
Parameter SHA-256: 526dea2d55d748caed94734139dda66064e194f74f46eee9b0f511ead38b7d1

XREAC / Water-cluster reference validation

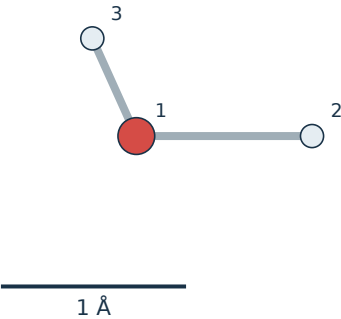
Monomer

3 atoms · O in red, H in pale gray · labels follow XYZ atom order · lines: bond order > 0.3

XY projection



Tilted projection (55° about x)

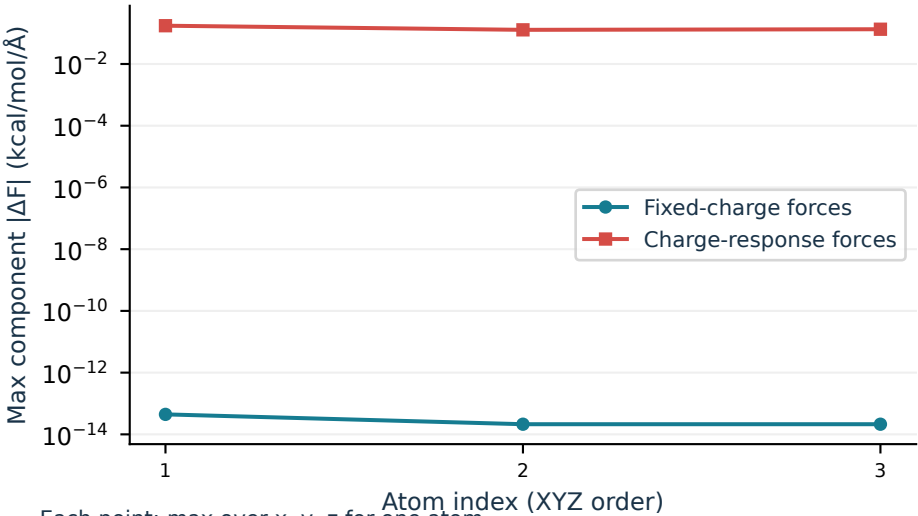


Quantity	Value
Python E (kcal/mol)	-248.344209206337
LAMMPS E (kcal/mol)	-248.344209206337
ΔE (kcal/mol)	+1.421085e-13
Fixed-charge max $ \Delta F $	4.440892e-14
Fixed-charge RMS ΔF	2.207580e-14
Charge-response max $ \Delta F $	1.739151e-01
Charge-response RMS ΔF	9.678950e-02

ENERGY COMPONENTS / kcal/mol

Component	Python	Δ vs LAMMPS
bond	-256.95789505	+0.00e+00
atom	-9.48962027	+0.00e+00
lone pair	-0.00000000	+0.00e+00
molecule	0.00000000	+0.00e+00
angle	3.08139970	+0.00e+00
penalty	0.00000000	+0.00e+00
angle conjugation	0.00000000	+0.00e+00
hydrogen bond	0.00000000	+0.00e+00
torsion	0.00000000	+0.00e+00
conjugation	0.00000000	+0.00e+00
vdw	48.67028283	+7.11e-15
coulomb	-78.87661327	+9.95e-14
electric field	0.00000000	+0.00e+00
qeq	45.22823685	+4.26e-14

Force differences from LAMMPS



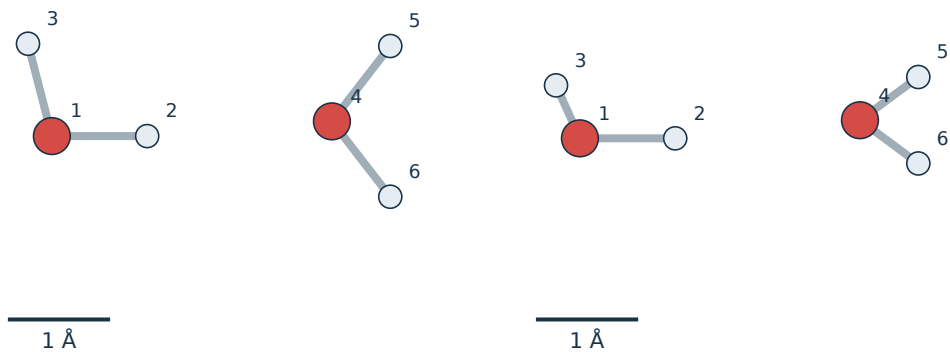
Each point: max over x, y, z for one atom.
Log scale; values below 10^{-16} shown at 10^{-16} .
Force summary above uses kcal/mol/Å throughout.

Dimer

6 atoms · O in red, H in pale gray · labels follow XYZ atom order · lines: bond order > 0.3

XY projection

Tilted projection (55° about x)

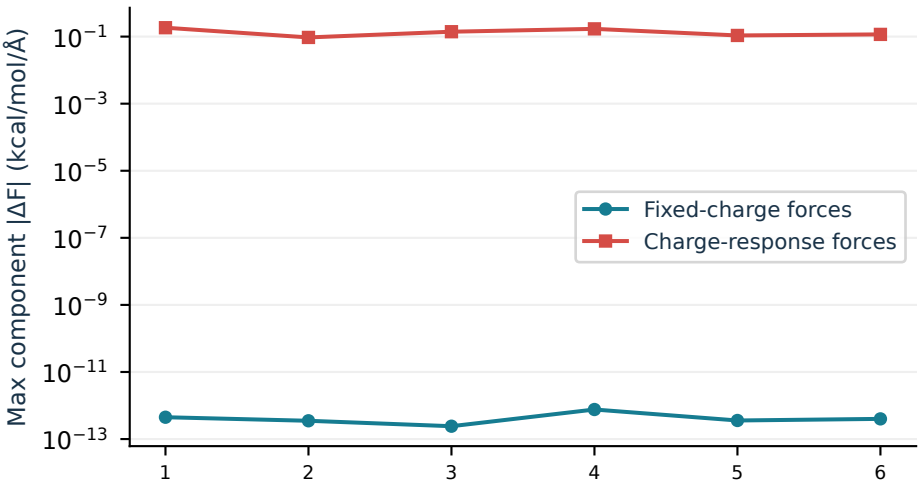


Quantity	Value
Python E (kcal/mol)	-502.630151520854
LAMMPS E (kcal/mol)	-502.630151520854
ΔE (kcal/mol)	+4.547474e-13
Fixed-charge max $ \Delta F $	7.602807e-13
Fixed-charge RMS ΔF	2.832414e-13
Charge-response max $ \Delta F $	1.840541e-01
Charge-response RMS ΔF	9.431702e-02

ENERGY COMPONENTS / kcal/mol

Component	Python	Δ vs LAMMPS
bond	-517.86751930	-1.14e-13
atom	-18.34985417	+0.00e+00
lone pair	-0.00000000	+4.14e-25
molecule	0.00000000	+0.00e+00
angle	6.86436142	+8.88e-16
penalty	0.00000000	+0.00e+00
angle conjugation	0.00000000	+0.00e+00
hydrogen bond	-1.96684546	+4.44e-16
torsion	0.00121985	+2.17e-18
conjugation	0.00000000	+0.00e+00
vdw	100.62557107	+1.42e-14
coulomb	-186.83832572	+1.85e-12
electric field	0.00000000	+0.00e+00
qeq	114.90124079	-1.53e-12

Force differences from LAMMPS



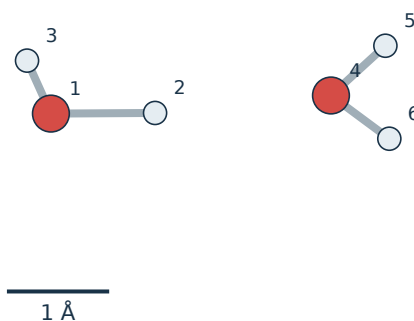
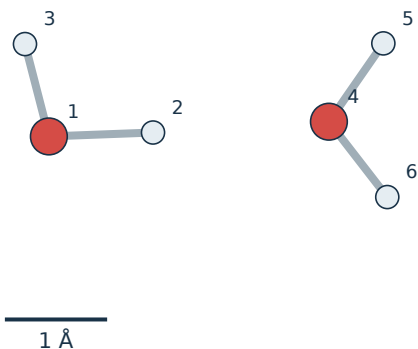
Each point: max over x, y, z for one atom.
Log scale; values below 10^{-16} shown at 10^{-16} .
Force summary above uses kcal/mol/Å throughout.

Distorted Dimer

6 atoms · O in red, H in pale gray · labels follow XYZ atom order · lines: bond order > 0.3

XY projection

Tilted projection (55° about x)

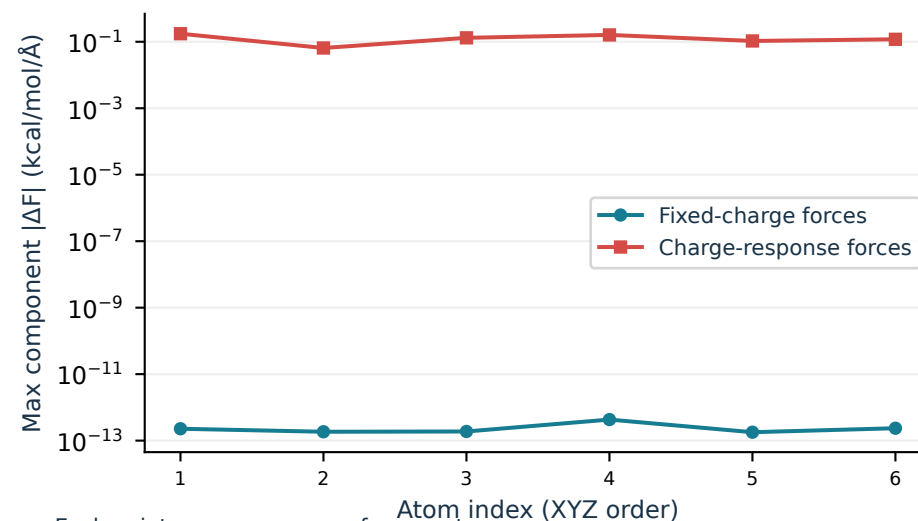


Quantity	Value
Python E (kcal/mol)	-498.807203051698
LAMMPS E (kcal/mol)	-498.807203051698
ΔE (kcal/mol)	+2.842171e-13
Fixed-charge max $ \Delta F $	4.294343e-13
Fixed-charge RMS ΔF	1.614888e-13
Charge-response max $ \Delta F $	1.747320e-01
Charge-response RMS ΔF	8.896202e-02

ENERGY COMPONENTS / kcal/mol

Component	Python	Δ vs LAMMPS
bond	-507.79364459	+0.00e+00
atom	-17.85610067	+0.00e+00
lone pair	-0.00000001	+0.00e+00
molecule	0.00000000	+0.00e+00
angle	7.00067788	+1.78e-15
penalty	0.00000000	+0.00e+00
angle conjugation	0.00000000	+0.00e+00
hydrogen bond	-1.45319620	-2.22e-16
torsion	0.00199824	+1.65e-17
conjugation	0.00000000	+0.00e+00
vdw	92.27169959	+0.00e+00
coulomb	-181.08981660	+1.28e-12
electric field	0.00000000	+0.00e+00
qeq	110.11117930	-9.95e-13

Force differences from LAMMPS



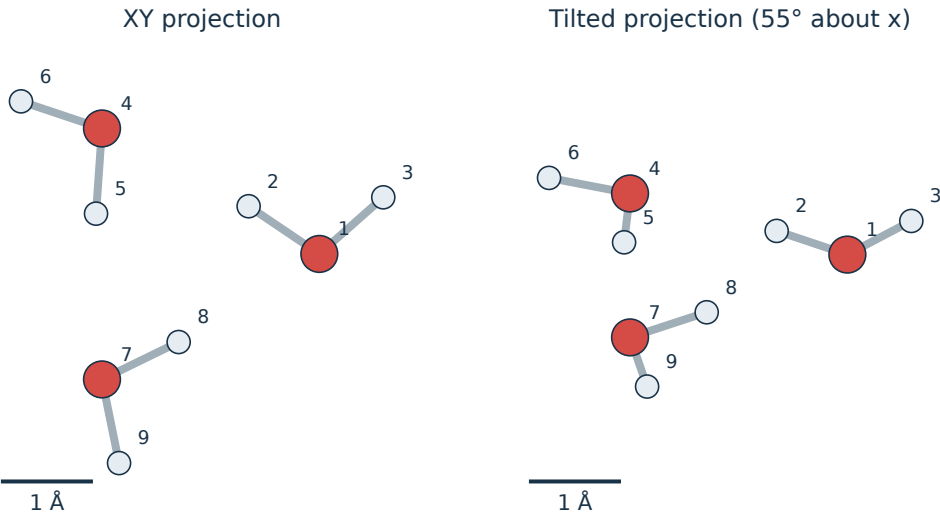
Each point: max over x, y, z for one atom.

Log scale; values below 10^{-16} shown at 10^{-16} .

Force summary above uses kcal/mol/Å throughout.

Trimer

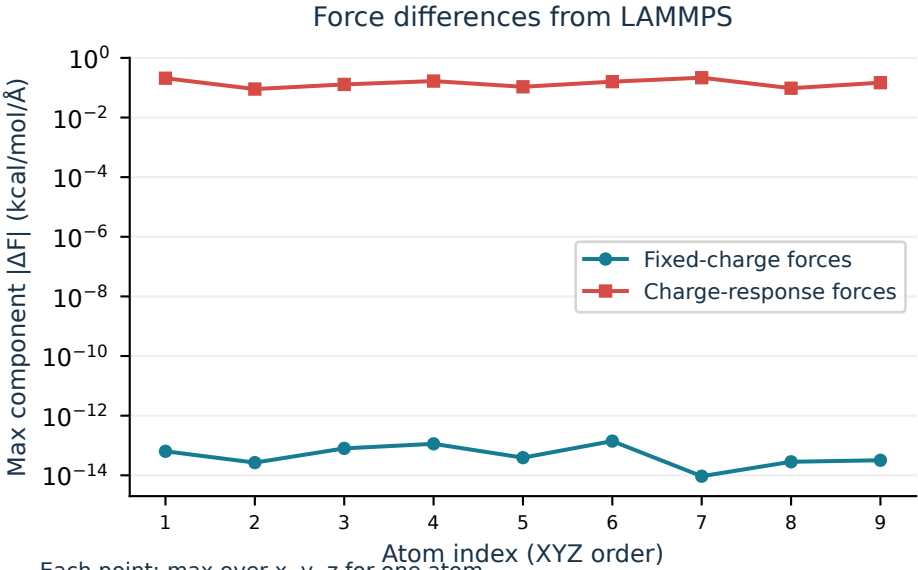
9 atoms · O in red, H in pale gray · labels follow XYZ atom order · lines: bond order > 0.3



Quantity	Value
Python E (kcal/mol)	-756.928700872728
LAMMPS E (kcal/mol)	-756.928700872729
ΔE (kcal/mol)	+4.547474e-13
Fixed-charge max $ \Delta F $	1.403322e-13
Fixed-charge RMS ΔF	4.427622e-14
Charge-response max $ \Delta F $	2.164235e-01
Charge-response RMS ΔF	9.777385e-02

ENERGY COMPONENTS / kcal/mol

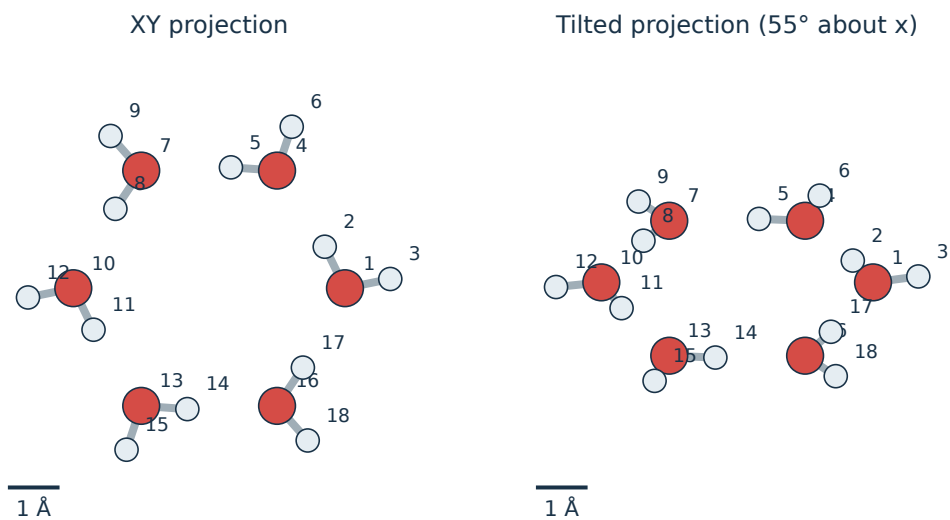
Component	Python	Δ vs LAMMPS
bond	-782.26544675	-1.14e-13
atom	-26.68402275	+3.55e-15
lone pair	-0.00000000	+0.00e+00
molecule	0.00000000	+0.00e+00
angle	12.29523602	+1.78e-15
penalty	0.00000000	+0.00e+00
angle conjugation	0.00000000	+0.00e+00
hydrogen bond	-7.00514051	-8.88e-16
torsion	0.00268766	-1.13e-17
conjugation	0.00000000	+0.00e+00
vdw	158.96026608	+0.00e+00
coulomb	-304.80692272	+2.27e-13
electric field	0.00000000	+0.00e+00
qeq	192.57464210	+2.84e-13



Each point: max over x, y, z for one atom.
Log scale; values below 10⁻¹⁶ shown at 10⁻¹⁶.
Force summary above uses kcal/mol/Å throughout.

Hexamer

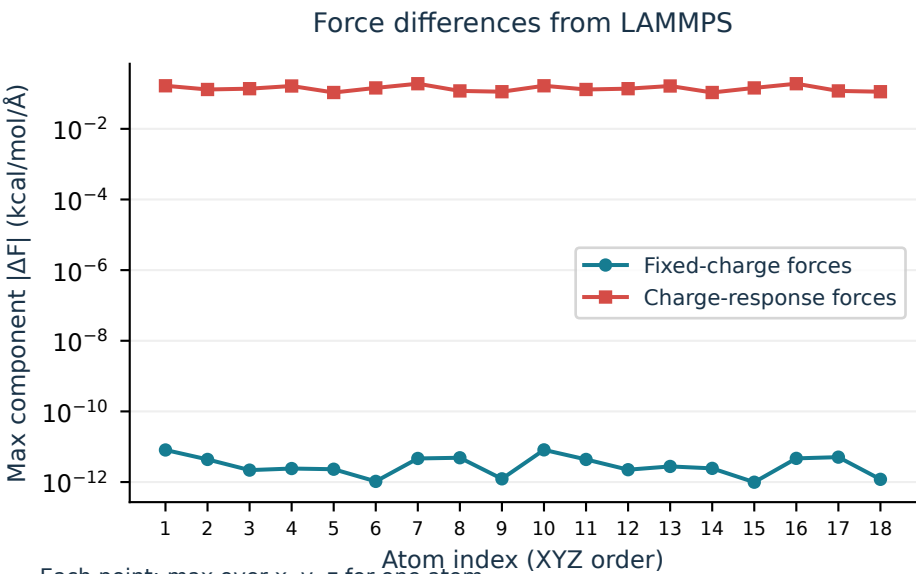
18 atoms · O in red, H in pale gray · labels follow XYZ atom order · lines: bond order > 0.3



Quantity	Value
Python E (kcal/mol)	-1524.961058062462
LAMMPS E (kcal/mol)	-1524.961058062461
ΔE (kcal/mol)	-6.821210e-13
Fixed-charge max $ \Delta F $	8.151257e-12
Fixed-charge RMS ΔF	2.566117e-12
Charge-response max $ \Delta F $	1.894887e-01
Charge-response RMS ΔF	9.102629e-02

ENERGY COMPONENTS / kcal/mol

Component	Python	Δ vs LAMMPS
bond	-1565.80037034	+0.00e+00
atom	-53.25895762	+0.00e+00
lone pair	-0.00000000	+0.00e+00
molecule	0.00000000	+0.00e+00
angle	22.62167228	-3.55e-15
penalty	0.00000000	+0.00e+00
angle conjugation	0.00000000	+0.00e+00
hydrogen bond	-11.68169530	+3.55e-15
torsion	0.00597983	-3.47e-18
conjugation	0.00000000	+0.00e+00
vdw	311.55513623	-2.27e-13
coulomb	-643.02829756	+1.59e-12
electric field	0.00000000	+0.00e+00
qeq	414.62547443	-5.12e-13



Each point: max over x, y, z for one atom.
Log scale; values below 10^{-16} shown at 10^{-16} .
Force summary above uses kcal/mol/Å throughout.